Literature search, screening and source verification

Compilation of life-history parameters of penaeid shrimps (Decapoda: Penaeidae) from the Brazilian coast

Methods document — version 3 (English edition)

14 October 2026

Graduate Program in Fisheries Resources and Aquaculture (PGRPAq)

Universidade Federal Rural de Pernambuco

This document serves as the basis for the Methods section of the manuscript and as a supplement for reproducibility.

Contents

1. Scope, nature of the synthesis and reporting standard
2. Eligibility criteria
3. Information sources
4. Search strategy
5. Record management, deduplication and screening
6. Validation of the search strategy
7. Metadata verification
8. Search results
9. Extraction, harmonization and curation
10. Reliability and non-independence among estimates
11. Statistical analysis plan
12. Current state of the extraction database
13. Limitations
14. Reproducibility and availability

Methodological references
Appendix A — Search log
Appendix B — Records not retrieved
Appendix C — List of sources and outcomes
Appendix D — Search strategy per database and execution record

1. Scope, nature of the synthesis and reporting standard

1.1. Objective

This document describes, in an auditable manner, the literature search strategy, the eligibility criteria, the screening procedure and the metadata verification adopted in the compilation of life-history parameters of commercially exploited penaeid shrimps on the Brazilian coast. The product of the compilation is an estimate-level database in which each row corresponds to one parameter value as reported in the original source, together with its unit, the sex to which it refers, the estimation method, the locality and the period of data collection.

1.2. Nature of the synthesis and terminology

Formally, the work is not an effect-size meta-analysis, because the compiled parameters do not share an effect metric with comparable variance across studies. It combines three recognized components: (i) a systematic map, whose result is a description of the coverage of knowledge by species, parameter, region and period; (ii) a trait-data compilation, whose product is the database that feeds the subsequent analyses; and (iii) a quantitative synthesis by mixed models (Section 11), whose object is not a weighted mean effect but the partitioning of the variance of the parameters among species, regions, research groups, methods and populations.

The method follows the guidelines of the Collaboration for Environmental Evidence (CEE Guidelines and Standards for Evidence Synthesis in Environmental Management, v5.0, 2022) and the systematic mapping procedures described by James, Randall & Haddaway (2016). Reporting follows PRISMA 2020 (Page et al. 2021), complemented by the PRISMA-EcoEvo extension (O'Dea et al. 2021), which is the standard applicable to syntheses in ecology and evolution and which explicitly addresses the problem of non-independence among estimates — a central issue in this body of literature.

Terminology. The work is reported as a "systematic map and quantitative synthesis" (or "systematic compilation of life-history parameters"). The labels "systematic review" and "meta-analysis" are not used, because they imply an effect question, a per-study risk-of-bias assessment and variance weighting, none of which is part of the design. The adjective "systematic" qualifies the search and the screening, which are systematic and reported to an international standard; it does not imply a meta-analysis. The search followed CEE v5.0 and is reported according to PRISMA 2020 and the PRISMA-EcoEvo extension, with the adaptations listed in Table 1.

Deliberate distinction: PRISMA is a reporting standard, not a method. It determines what must be counted and declared; the methodological design derives from CEE v5.0.

1.3. Statement on the timing of protocol registration

Data extraction began in August 2026, before the search protocol was formalized. This circumstance is declared explicitly, in line with item 3 of the PRISMA-EcoEvo checklist. The protocol will be deposited in a public repository (OSF) with the actual date of registration and with a statement that no eligibility criterion was changed after extraction began. The studies processed before registration were fully reprocessed under the same criteria and became part of the reference set (benchmark set) used to assess the sensitivity of the search strategy (Section 6).

1.4. Adherence to PRISMA 2020 and adaptations

PRISMA 2020 (Page et al. 2021) is a reporting standard: a 27-item checklist and a flow diagram that require authors to declare what was searched, how, where, how much was found and what was done with each record. It was designed for reviews that estimate intervention effects, so some items (risk of bias, effect measures, certainty of evidence) have no direct counterpart in a compilation of parameters. The PRISMA-EcoEvo extension (O'Dea et al. 2021) adapts the checklist to ecology and evolution, and PRISMA-S (Rethlefsen et al. 2021) details the reporting of the search. Adopting PRISMA does not require abandoning the planned statistical analyses: the standard governs reporting, not the model. Table 1 shows where each item is reported in this document and the adaptations made.

Table 1. Location of the PRISMA 2020, PRISMA-EcoEvo and PRISMA-S items in this document, with adaptations.

Item(s)	Topic	Where reported	Adaptation / note
1–2	Title and abstract	Manuscript title and abstract	The title identifies the design ("systematic map and quantitative synthesis"); structured abstract according to PRISMA for Abstracts.
3–4	Rationale and objectives	Sections 1.1, 1.2 and 11.1	The confirmatory hypotheses are fixed in Section 11.1 (Table 8) and registered before the models are run.
5	Eligibility criteria	Section 2	None.
6–7	Information sources and search strategy (PRISMA-S)	Sections 3, 4, Appendices A and D	The same string was used in all databases. Appendix D gives the adaptation to the syntax of each database and the execution record (Table D2), with the date of execution and the year coverage of each database.

Item(s)	Topic	Where reported	Adaptation / note
8	Selection process	Section 5.3	Screening and reading by two researchers (M.S.S. and A.C.A.F.), with independent comparison of the results and re-reading of the article when they diverged. Agreement is reported as the number of divergent sources, without Cohen's κ.
9–10	Data collection process and data items	Sections 9.1–9.2	Data dictionary in Section 9. Extraction by two researchers, validated with a language model and compared independently (Section 9.2).
11, 18	Risk of bias in studies	Section 10	Adapted: there is no standard risk-of-bias instrument for trait estimates. It is replaced by a 1–5 reliability scale, the definition of the length measure (cl_definition) and quality-control (QC) flags. The adaptation is justified in the manuscript.
12–13, 20	Effect measures and synthesis methods	Section 11	Adapted: there is no effect size. The synthesis uses Bayesian multilevel (mixed) models; the analysis plan is deposited before the analysis.
14, 21	Reporting bias	Section 11.5	Adapted: funnel plots do not apply. Reporting bias is addressed by sensitivity analyses by publication type, language and grey literature.
15, 22	Certainty of evidence	Not applicable	GRADE assumes an effect question. The omission is declared, not omitted silently.
16–17	Study selection and study characteristics	Section 8, Figure 1, Table 6 and Section 12	Flow diagram (Figure 1, generated by script) and counts per stage (Table 6, Section 8); source characteristics in Section 12.
24	Registration and protocol	Section 1.3	Deposit at OSF with the actual date of registration and a statement that extraction preceded the protocol.
25–27	Support, conflicts of interest and data availability	Section 14	Database with DOI and CC-BY license; public code and analysis plan.
PRISMA-EcoEvo: non-independence, harmonization of units, heterogeneity, sensitivity	Non-independence and heterogeneity among estimates	Sections 9–11	Identification columns were added to the master spreadsheet on 14 October 2026 (Section 10.2).
Use of computational assistance in extraction	Computational assistance	Section 9.2	Declared in Section 9.2.

The work is reported according to PRISMA 2020 and PRISMA-EcoEvo, with the adaptations justified in Table 1. Full adherence is not claimed, because the items on risk of bias, effect measures and certainty of evidence do not apply without adaptation.

1.5. Document history

- **Version 1 (8 Sep 2026).** Scope, eligibility criteria, sources, search strategy, screening of 188 records, validation and metadata verification.
- **Version 2 (14 Oct 2026).** Terminology and PRISMA adherence table (Sections 1.2 and 1.4); correction of the treatment of captive studies and of the macro-region count (Sections 2.1 and 2.3); extraction, harmonization and curation (Section 9); reliability and non-independence (Section 10); statistical analysis plan (Section 11); current status of the database (Section 12); expanded limitations (Section 13); per-database search strings and execution record (Appendix D); additional references.
- **Version 3 (14 Oct 2026).** Searches of Web of Science, Scopus, ASFA and the CAPES Periodicals Portal and citation searching (Sections 3.2 and 3.3); PRISMA flow diagram and counts (Section 8, Table 6); screening and extraction by two researchers with independent comparison (Sections 5.3 and 9.2); identification and redundancy-control columns added to the master spreadsheet (Section 10.2); hypotheses aligned with the OSF analysis plan (Table 8); removal of one study outside Brazilian waters (Table 7); update of all database counts (Section 12). English edition prepared 14 Oct 2026.

2. Eligibility criteria

The criteria below were fixed in writing before the search described in Section 4 was run and apply to all records, regardless of publication type.

2.1. Population

Studies are eligible if they report life-history parameters for natural populations of the following Penaeidae species of commercial importance in Brazil:

Accepted name (WoRMS)	Common name	Synonyms and spellings retrieved by the search
Penaeus (Xiphopenaeus) kroyeri (Heller, 1862)	seabob shrimp (camarão sete-barbas)	*Xiphopenaeus kroyeri*; sea-bob; seabob; Atlantic seabob
Xiphopenaeus dincao Carvalho-Batista et al., 2019	seabob shrimp (camarão sete-barbas)	species described in 2019; earlier records from the Southeast and South attributed to *X. kroyeri* may correspond to this species or to a mixture of the two
Penaeus (Farfantepenaeus) subtilis (Pérez Farfante, 1967)	pink shrimp (camarão-rosa), camarão-legítimo	*Farfantepenaeus subtilis*; *Penaeus subtilis*; southern brown shrimp
Penaeus (Farfantepenaeus) brasiliensis Latreille, 1817	pink shrimp (camarão-rosa)	*Farfantepenaeus brasiliensis*; *Penaeus brasiliensis*; pink shrimp
Penaeus (Farfantepenaeus) paulensis (Pérez Farfante, 1967)	pink shrimp (camarão-rosa)	*Farfantepenaeus paulensis*; *Penaeus paulensis*
Penaeus (Litopenaeus) schmitti (Burkenroad, 1936)	white shrimp (camarão-branco)	*Litopenaeus schmitti*; *Penaeus schmitti*; white shrimp; southern white shrimp

Taxonomic note, decisive for the recall of the search: WoRMS has downgraded Farfantepenaeus, Litopenaeus and Fenneropenaeus to subgenera of Penaeus. The Brazilian literature uses all three nomenclature schemes depending on the decade of publication. Searching under a single generic scheme eliminates entire decades of records. For this reason, all searches were conducted on the specific epithet, decoupled from the genus (for example, "kroyeri", "subtilis", "schmitti"), and not on the full binomial.

In the current version of the master spreadsheet, all estimates for the genus *Xiphopenaeus* are listed under *X. kroyeri*, the name as published; no row is attributed to *X. dincao*. Studies published after 2019 that recognize both species are recorded with the accepted name and the AphiaID in separate fields, and the taxonomic uncertainty of earlier records is declared as a limitation (Section 11.5).

Studies conducted in captivity, in the laboratory or in culture are not eligible for the database of natural populations. The records of this nature identified during extraction (two studies, 18 rows) were removed from the master spreadsheet and preserved, with the reason, in the removed_rows sheet (Section 9.3) for a possible sensitivity analysis. Studies on other penaeid species are eligible only when they report, in the same work, parameters for at least one of the six species above.

2.2. Outcomes (parameters of interest)

A study is eligible if it reports, for at least one of the target species, at least one of the parameters below as a numerical value of the authors' own. Studies that only cite values from third parties are handled as described in Section 2.6.

Domain	Parameters
Growth	Asymptotic length ($L\infty$, in carapace length or total length); growth coefficient (k); time-position parameter (t_0); asymptotic weight ($W\infty$); parameters C and WP of the seasonalized model; growth performance index φ'
Longevity	Estimated longevity (t_max) and the criterion used (4.605/k, 3/k, tagging, other)
Mortality	Natural mortality (M); total mortality (Z); fishing mortality (F); exploitation rate (E); reference point E_MSY; estimation method for each
Maturation	Length at first gonadal maturity (L50) by sex, in CL and TL; length at first sexual/morphological maturity (SL50); stage classification criterion
Reproduction and recruitment	Reproductive period (start and end); recruitment period (start and end); recruitment type declared by the authors
Biometric relationships	Weight-length relationship (a and b of the power function); relationship between total length and carapace length (a and b) and the functional form adopted
Sampling structure	Sample size (n); minimum and maximum observed lengths by sex; length at recruitment (Lr)

In all cases, the unit in which the parameter was reported in the study is recorded, without conversion in the raw-value cell. The conversions required for analysis are executed in a versioned script, in derived columns, so that they remain auditable and correctable.

2.3. Geographic scope

Studies are eligible if the data collection area lies within Brazilian jurisdictional waters, including estuaries, the coastal region and the continental shelf, of the four coastal macro-regions (North, Northeast, Southeast and South) and the Exclusive Economic Zone. Studies conducted in other countries are excluded, even when on the same

species, because they represent distinct populations and exploitation regimes; the decision is recorded in the protocol and the list of these records is preserved for a possible sensitivity analysis.

2.4. Publication types

There is no restriction by publication type. Eligible items are articles in peer-reviewed journals, PhD theses, MSc dissertations, undergraduate theses, institutional technical-scientific bulletins, government technical reports, fisheries management plans, book chapters and full papers published in conference proceedings.

The unrestricted inclusion of grey literature is not a liberality: in tropical fisheries, and particularly for penaeids in Brazil, a substantial fraction of the growth and mortality estimates exists exclusively in institutional bulletins and in theses not published as articles. Excluding this literature would produce a systematically biased picture of the coverage of knowledge — which is precisely one of the outcomes of the systematic map. Of the 220 records identified, 32 (14.5%) were retrieved by grey-literature searching (Section 8).

2.5. Language and period

There is no language restriction. Searches were conducted in Portuguese, English and Spanish. Restricting the search to English-language publications, on a topic whose primary literature is mostly in Portuguese, would constitute language bias and compromise the validity of the synthesis.

There is no lower or upper time limit. The oldest estimates of mortality and of the stock-recruitment relationship for *Xiphopenaeus kroyeri* in Brazil date from 1971 (Santos, Neiva & Valentini, Boletim do Instituto de Pesca 1(1)), and a date cutoff would eliminate them. The search therefore covers the period from 1971 to the date of execution; the databases were searched from their inception.

2.6. Exclusion criteria and handling of secondary citations

The following are excluded, with the record identified by name and the reason categorized: (i) studies that report none of the parameters of interest; (ii) studies on species outside the scope; (iii) studies conducted outside Brazilian waters; (iv) studies exclusively on aquaculture, fishing technology, processing, socioeconomics or contamination, without a biological parameter; (v) conference abstracts without retrievable numerical data.

A value located only as a secondary citation (apud) is not incorporated into the database unless the original work is located and read. In the current database, 57 estimates (4.3% of rows), from 19 sources, are of secondary origin. When the original is inaccessible, a pre-specified rule applies: the value is kept, flagged with its reliability level and linked to the identifier of the primary study, so that it is not treated as an independent observation. When two or more secondary sources attribute divergent values to the same inaccessible original, all are recorded, flagged by source, with no arbitration among them.

This is the main pseudoreplication risk of the dataset: the same value entering the analysis three or four times through three or four distinct secondary sources. The control is structural, not editorial. In the current version, the master spreadsheet carries the source read (source_read_id), the population sampled (population_id; 184 populations) and the research group; the identifiers of the estimate (estimate_id) and of the primary study (primary_study_id) and the reliability level (reliability_level) were added before any analysis (Section 10.2).

3. Information sources

3.1. Sources and search timeline

The search was carried out by two researchers: Matheus Lourenço Soares da Silva (M.S.S.; ORCID https://orcid.org/0000-0001-6382-8698) and Anna Carolina Arnaldo de Figueiredo (A.C.A.F.; ORCID https://orcid.org/0000-0003-2569-972X), and comprised four components:

- (i) Database searches: Web of Science Core Collection, Scopus, ASFA and the CAPES Periodicals Portal, all searched with the same string (Section 4.1 and Appendix D).
- (ii) Other methods: direct (forward) and reverse (backward) citation searching, grey-literature searching and complementary searching.
- (iii) AI-assisted programmatic web retrieval (the "complementary search"), carried out on 7–8 September 2026, the only searches of that month (none were run between 9 September and early October), and updated on 12–13 October 2026, by programmatic access and direct browsing, over the sources listed in Appendix A. The sources with the highest yield were, in this order: the Boletim do Instituto de Pesca index, the Crossref REST API, the CEPENE/ICMBio bulletin portal and the institutional repositories of USP, UNESP and UFPA.
- (iv) Final update searches, in early October 2026, to retrieve the remaining articles.

The timeline of the searches, at month level, is given in Table 2. The entry dates in the master spreadsheet and the creation dates of the individual per-article spreadsheets document the collection dates.

Table 2. Timeline of the searches (month level).

Period	Searcher	Search component	Sources
Jul 2026	A.C.A.F.	Database searches and other methods (mainly direct and reverse citation searching); about three sessions per week; covered all sources	WoS, Scopus, ASFA, CAPES Periodicals Portal; citation chains; grey literature; repositories
Aug 2026	M.S.S.	Database searches and other methods (mainly direct and reverse citation searching); about three sessions per week; covered all sources	WoS, Scopus, ASFA, CAPES Periodicals Portal; citation chains; grey literature; repositories
Sep 2026 (7–8 Sep)	M.S.S.	AI-assisted programmatic web retrieval (complementary search) and further searches on 7–8 Sep; no searches between 9 Sep and early Oct	Appendix A sources
Early Oct 2026	M.S.S.	Two final update sessions (databases and other methods) to retrieve remaining articles; assisted retrieval re-run 12–13 Oct	Databases; other methods; Appendix A
Jul–Oct 2026	A.C.A.F. and M.S.S.	Independent comparison of the two search and extraction results, covering all sources	All

Each source consulted is recorded in Appendix A with the access mode, the outcome of the query and the pertinent remark, including the sources that could not be consulted and the technical reason. The record of failures is part of the method: a declared inaccessible source is a delimited limitation; an omitted inaccessible source is a gap that the reader cannot assess.

3.2. Bibliographic databases and sources not searched

The four bibliographic databases were searched manually by the researchers, outside the assisted retrieval described in Section 3.1, with the same search string in all databases (Section 4.1 and Appendix D): Web of Science (Core Collection), Scopus, ASFA and the CAPES Periodicals Portal. Coverage extended from the inception of each database to October 2026. The records returned and those considered relevant by title and abstract were as follows (Table D2):

- **Web of Science, Core Collection:** 24 records returned, of which 20 relevant.
- **Scopus:** 36 records returned, of which 30 relevant.
- **ASFA (Aquatic Sciences and Fisheries Abstracts):** 6 records returned, of which 5 relevant. It is the database that indexes national bulletins and technical fisheries reports not covered by Web of Science and Scopus.
- **CAPES Periodicals Portal:** 6 records returned, of which 5 relevant.

The following sources were not searched. The omission is declared as a limitation (Section 13):

- Zoological Record and BIOSIS Previews.
- LILACS, for Latin American literature in health and related sciences not covered by SciELO.
- Google Scholar, for which a stop rule was defined (the first 200 references ordered by relevance, screened by title) so that the query would be reproducible; it was not run.

SciELO Brasil and BDTD/IBICT, although not accessible programmatically in the assisted retrieval (Appendix A), had their content partially retrieved indirectly, through Crossref and DOAJ, which index them, and by direct access to the institutional repositories. This indirect coverage is imperfect and does not substitute for searching the native interfaces, and this limitation is declared in Section 13.

3.3. Citation searching

In addition to the database searches, citation searching was carried out in two directions, formalized as a method and not left to chance:

- Reverse (backward): full scan of the reference list of each study already included, with named identification of every source that reports a parameter of interest. This route identified a good share of the pending records listed in Appendix B, among them the original sources of methodological criteria used recurrently in the Brazilian literature.
- Forward: identification of the works that cite each included study.

Citation searching, in both directions, identified 30 records, whose texts were sought individually.

When the original of a cited source could not be located, the values were read from a work that cites it (secondary citation, or apud), recorded in the master spreadsheet with source_read_id and reliability_level 4 (Section 10.3). In

the current database there are 19 sources and 57 estimates in this condition, of which 17 sources belong to the initial compilations and 2 came from the search.

4. Search strategy

4.1. Structure of the strategy

The strategy was built in two blocks combined by conjunction — population and outcome — deliberately without a third block for study design, whose inclusion reduces sensitivity without appreciable gain in precision in trait syntheses. A third, geographic block was used only as an auxiliary filter in high-yield queries and never as a mandatory restriction, so as not to eliminate Brazilian studies that do not name the country in the title or abstract.

Block A — population

```
"kroyeri" OR "dincao" OR "subtilis" OR "brasiliensis" OR "paulensis" OR
"schmitti" OR "Xiphopenaeus" OR "Farfantepenaeus" OR "Litopenaeus" OR
"Penaeus" OR "Penaeidae" OR "penaeid*" OR "peneide*" OR "peneido*" OR
"shrimp*" OR "prawn*" OR "camarão" OR "camarões" OR "camaron*" OR
"sete-barbas" OR "sete barbas" OR "seabob" OR "sea-bob" OR
"camarão-rosa" OR "camarão-branco" OR "pink shrimp" OR "white shrimp" OR
"brown shrimp" OR "camarão-legítimo"
```

Block B — outcome

```
"von Bertalanffy" OR "growth parameter*" OR "growth curve" OR "crescimento" OR
"asymptotic length" OR "L-infinity" OR "Linf" OR "L∞" OR "comprimento assintótico" OR
"growth coefficient" OR "ELEFAN" OR "FiSAT" OR "TropFishR" OR
"length-frequency" OR "frequência de comprimento" OR "progressão modal" OR
"natural mortality" OR "total mortality" OR "fishing mortality" OR "mortalidade" OR
"exploitation rate" OR "taxa de explotação" OR "stock assessment" OR
"avaliação de estoque" OR "dinâmica populacional" OR "population dynamic*" OR
"size at maturity" OR "length at maturity" OR "L50" OR "LM50" OR "CL50" OR
"maturidade" OR "maturação" OR "primeira maturação" OR "madurez" OR
"length-weight relation*" OR "relação peso-comprimento" OR "relações biométricas" OR
"biologia reprodutiva" OR "reproductive biology" OR "período reprodutivo" OR
"recrutamento" OR "recruitment" OR "longevity" OR "longevidade" OR
"fecundidade" OR "fecundity" OR "razão sexual" OR "sex ratio" OR
"estrutura populacional" OR "biologia populacional" OR "defeso"
```

Block C — geographic (auxiliary filter, never mandatory)

```
"Brazil" OR "Brasil" OR "Brazilian" OR "Amazon* shelf" OR "plataforma continental" OR
state and locality names: Pará, Maranhão, Ceará, Pernambuco, Sergipe, Bahia,
Espírito Santo, Rio de Janeiro, São Paulo, Paraná, Santa Catarina, Rio Grande do Sul,
Ubatuba, Cananéia, Santos, Baixada Santista, Macaé, Atafona, Lagoa dos Patos,
Sirinhaém, Tamandaré, Caravelas, Corumbau, Curuçá, Sepetiba
```

4.2. Adaptation of the strategy to each interface

None of the interfaces actually accessible to the assisted retrieval accepts the full canonical string. The strategy was therefore decomposed into smaller queries, always preserving the A ∧ B structure. The adaptations are described below. These adaptations are reported in Appendix D.

Interface	Adaptation adopted
Crossref REST API	Queries to the /works endpoint with the query.bibliographic parameter receiving pairs of terms (one from block A, one from block B), rows=40, sorted by relevance. No support for explicit Boolean operators or truncation: decomposition into pairs is mandatory. Covers SciELO Brasil, Boletim do Instituto de Pesca, Nauplius, Arquivos de Ciências do Mar, Atlântica, Revista Brasileira de Zoologia and USP theses (DOI prefix 10.11606).
DOAJ API	Endpoint /api/search/articles with the same pairs of terms. High yield for Brazilian open-access journals.
OJS portal of the Boletim do Instituto de Pesca	Browsing and searching of the index by species and by outcome term, complemented by a scan of the tables of contents of the issues. Pitfall recorded: the system assigns the year 2018 to almost all retro-digitized articles, so the publication year has to be read from the body of the record or from the PDF, never from the metadata field. This pitfall produced incorrect year assignments in the initial screening, which were subsequently corrected (Section 7).
CEPENE/ICMBio portal	Direct access to the issue and article PDFs. Volumes 12, 13(2), 14 (partial), 15(2), 16, 17 and 18 have an extractable text layer. Volumes 1 to 11, 13(1) and 15(1) are scanned without optical character recognition or only partially available, which makes them unrecoverable by text search — the origin of most of the pending records in Appendix B.
Institutional repositories	The search endpoints of the DSpace installations block automated access; individual files, when located by web search, open normally. The effective strategy was therefore to locate the record by web search restricted to the institution's domain and to access the bitstream directly.
General web search	Used for grey literature and as a workaround for sources with blocked programmatic access, always with the same pairs of terms and with domain restriction where pertinent.

4.3. Effective coverage of the execution

The search was conducted in four parallel and independent fronts, segmented by species and by parameter domain, so that each record had more than one possible route of retrieval. The fronts were consolidated into a single list, with deduplication by normalized title and by persistent identifier (DOI, handle or URL). Deduplication was verified automatically: after consolidation, no repeated title remained in the list, except for an intentional placeholder for an unidentified record.

Note: the interfaces of the assisted retrieval (REST APIs and browsing of OJS portals) do not generate exportable record sets with stable counts per query. This arm therefore enters the flow diagram as "complementary searching" (n = 86), corresponding to the 188 unique records of the consolidated list plus the 22 records excluded before consolidation and the 10 duplicates, less the 72 database records, the 30 citation records and the 32 grey-literature records. The per-database counts (Table D2) come from the search histories of each database.

5. Record management, deduplication and screening

5.1. Deduplication

Deduplication was performed with three successive keys: persistent identifier (DOI or handle), normalized title (lowercase, without accents and without punctuation) and author-year pair. Specific care was needed for pairs of works from the same group published in the same venue and year, which title normalization tends to merge improperly. Two cases were examined individually and kept as distinct records after verification of the identifier: Miazaki et al. (2016), Nauplius 24: e2016025, and Costa et al. (2016), Nauplius 24: e2016024.

Across the databases, citation searching, grey literature and complementary searching, 10 duplicate records were removed before screening (Figure 1).

5.2. Screening and prioritization

Screening was conducted on title, abstract and, when needed to decide, full text. In addition to the inclusion decision, each record received a priority class that reflects the expected yield of parameters and guides the order in which full texts are obtained and read. The priority class is a work-management instrument and not an eligibility criterion: no eligible record is discarded for low priority.

Class	Definition	Expected content
1 — highest	Compilations, synopses and stock assessments	Gather time series and multiple estimates in a single document; include national management plans, species synopses and research network reports
2 — high	Own estimates of growth and/or mortality	$L\infty$, k, t_0, M, Z, F, exploitation rate obtained by the authors themselves
3 — medium	Maturation, reproduction, recruitment, biometric relationships	L50, SL50, reproductive period, recruitment period, parameters of the length-weight relationship and TL×CL
4 — marginal	Size structure, sex ratio, abundance	Minimum and maximum lengths, n, frequency distribution; contribute to the denominator of the map and to the sampling-truncation covariates
0 — pending	Record identified, text not located	Existing and verified reference, with no accessible online version; see Appendix B
9 — already in database	Record retrieved that had already been extracted	Makes up the benchmark set for assessing the sensitivity of the search

5.3. Screening and reading by two researchers

The 210 records resulting from deduplication were screened by title and abstract by two researchers, M.S.S. and A.C.A.F. A total of 24 were excluded: 12 unrelated to the topic among the raw database returns, 10 outside Brazilian waters (not read) and 2 without a parameter of interest or of non-primary nature (a set of risk-assessment sheets and a histological study). Of the 186 reports sought, 13 were not retrieved in full text (11 not located and 2 records indicated as already in the initial database, with no match in the master spreadsheet) and 173 were read. In the reading, 62 reports were excluded: 55 without parameters of interest, 3 of species out of scope, 3 secondary compilations without primary data and 1 captivity study. A total of 111 reports from the search remained, to which 51 from the initial set of compilations are added, for 162 reports included.

6. Validation of the search strategy

The sensitivity of a search strategy is assessed against a benchmark set composed of works that are known to be retrievable. In this work, the benchmark set is formed by the studies already extracted before the protocol was formalized — which constitutes a benchmark independent of the strategy, since its assembly did not derive from it.

The search independently retrieved 34 records that were already in the initial database, without the strategy having been informed of their existence. None of these records required a later adjustment of the strategy to be retrieved. The formal sensitivity estimate on the full nominal list of the benchmark set (target ≥ 90%; terms of missing records extracted and the strategy iterated if below target) is reported in the manuscript in the form "the strategy retrieved x of y articles of the benchmark set (z% sensitivity); the missing ones were retrieved by citation searching".

A second test, of a different nature and already carried out, is the retrieval test of sources known only through citation. Several works were identified in this search from citations found in the studies already extracted and were then located independently in the databases — among them Heckler et al. (2013), Boletim do Instituto de Pesca 39(3): 283-297, and Almeida et al. (2012), Aquatic Biology 17(1): 57-69. The convergence of the two routes is evidence of the consistency of the strategy.

7. Metadata verification

All records in the list were subjected to metadata verification against the primary editorial source — the journal portal, the Crossref record or the published PDF — and not against third-party reference lists. The verification identified six discrepancies, of which three correspond to errors that circulate in the secondary literature and that, if propagated, would compromise the traceability of citations. The corrections are recorded by name in the references spreadsheet and reproduced below.

Record	Metadata before verification	Metadata after verification at the source
Santos, Neiva & Valentini — mortality curve of X. kroyeri	Year recorded as c. 1969; volume and issue unknown	1971; Bol. Inst. Pesca 1(1): 39-46. Verified in the journal index on 8 Sep 2026
Santos, Neiva & Valentini — reproduction curve of X. kroyeri	Year recorded as c. 1969; volume and issue unknown	1971; Bol. Inst. Pesca 1(1): 15-22. Same issue as the previous one
Growth and mortality of Penaeus subtilis on the Amazon continental shelf	Attributed to Aragão, Cintra et al., year and volume to be confirmed	Cintra, Aragão, Silva & Petrere Júnior (2021); Arq. Ciênc. Mar 54(1): 69-88; DOI 10.32360/acmar.v54i1.41667. The authorship had been incorrectly ordered in the initial screening
Severino-Rodrigues, Pita, Graça-Lopes, Coelho & Puzzi	Year 1993, according to the reference list of Sahm et al. (2011) and other secondary sources	1992; Bol. Inst. Pesca 19 (single issue): 67-81; sampling of 1979-1980. The year 1993 that circulates in the literature is incorrect
Ruas, Dumont & D'Incao (2011)	Pagination differing between sources: 127-138 or 161-172	Atlântica 33(2): 161-172; DOI 10.5088/atl.2011.33.2.161. Verified in the published PDF
Almeida, Baeza, Fransozo, Castilho & Fransozo (2012)	Ambiguous DOI: 10.3354/ab00462 or 10.3354/ab00469	10.3354/ab00462, confirmed in the Crossref record. The form ab00469 is incorrect
Santos, Severino-Rodrigues & Vaz-dos-Santos (2008)	Suspected duplication of a record already extracted	Bol. Inst. Pesca 34(3): 375-389, Baixada Santista. Identity confirmed with the record already in the initial database; reclassified as not new

Two systematic metadata pitfalls were identified and are declared here, because they affect any researcher who repeats this search. The first is the assignment of the year 2018 by the editorial system of the Boletim do Instituto de Pesca to retro-digitized articles from earlier decades. The second is the propagation, in the reference lists of peer-reviewed articles, of incorrect years and pagination for old works of the same journal — the Severino-Rodrigues et al. case is exemplary, and the incorrect year is replicated in multiple sources.

8. Search results

The consolidated list of the assisted retrieval comprised 188 unique records. At first consolidation, 130 were new material to obtain and process, 34 corresponded to studies already extracted and 24 were records whose existence was verified but whose full text was not located online. Tables 3 to 5 describe this initial composition (September 2026 consolidation of the assisted retrieval list). The final outcome of each stage, after full-text reading and incorporation of the database and citation searches, is in Table 6 and Figure 1.

Table 3. Distribution of screened records by priority class.

Class	n	% of total
1 — highest (compilations, synopses and stock assessments)	19	10.1
2 — high (original growth and/or mortality estimates)	12	6.4
3 — medium (maturation, reproduction, recruitment, biometrics)	56	29.8
4 — marginal (size structure, sex ratio, abundance)	43	22.9
Subtotal — new records to obtain	130	69.1
0 — pending (full text not located)	24	12.8
9 — already in the database	34	18.1
Total screened records	188	100.0

Table 4. New records by target species. A single record may cover more than one species, so the sum exceeds 130.

Species	New records
Penaeus (Xiphopenaeus) kroyeri and *Xiphopenaeus dincao*	60
Penaeus (Litopenaeus) schmitti	31
Penaeus (Farfantepenaeus) subtilis	20
Penaeus (Farfantepenaeus) paulensis	16
Penaeus (Farfantepenaeus) brasiliensis	11

Table 5. New records by publication type.

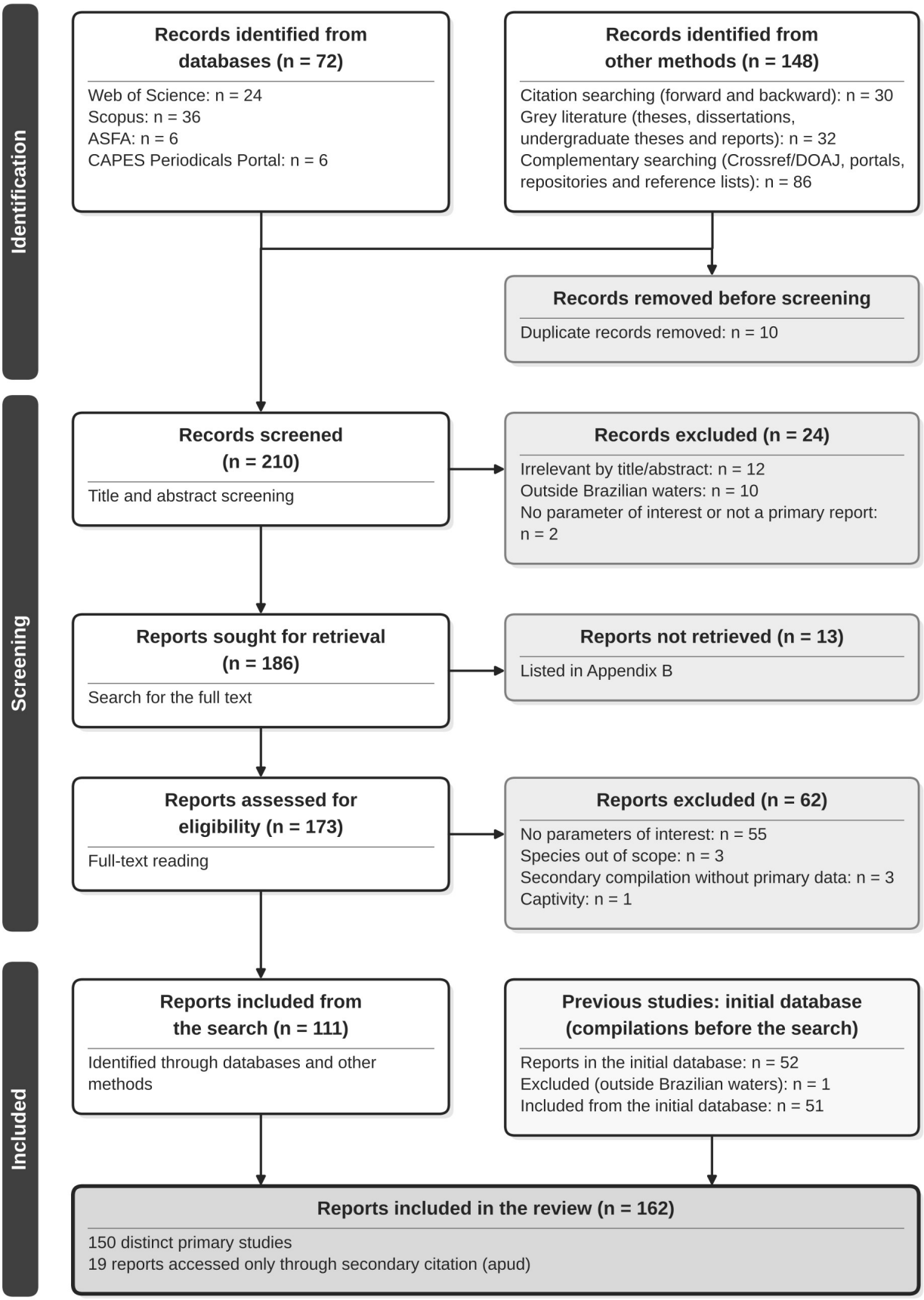
Publication type	n	% of 130
Journal article	106	81.5
PhD thesis	8	6.2
MSc dissertation	8	6.2
Technical report, management plan or book chapter	6	4.6
Undergraduate thesis	1	0.8
Full paper in conference proceedings	1	0.8
Total	130	100.0

The distribution by species reproduces a known pattern that is itself a result of the systematic map: research effort is strongly concentrated on *Xiphopenaeus kroyeri*, whereas *Farfantepenaeus brasiliensis*, despite its commercial relevance, has fewer than one-fifth as many records as the former. The distribution by publication type confirms the need for unrestricted inclusion of grey literature: 24 new records, almost one-fifth of the total, are not journal articles and would be invisible to a search restricted to commercial databases.

Table 6. Counts by stage of the PRISMA 2020 flow (Figure 1).

Phase	Step	n
Identification	Records from databases (Web of Science 24, Scopus 36, ASFA 6, CAPES 6)	72
	Records from other methods (citation searching 30, grey literature 32, complementary searching 86)	148
	Duplicates removed before screening	10
Screening	Records screened (title and abstract)	210
	Excluded at screening (irrelevant 12, outside Brazilian waters 10, no parameter or non-primary 2)	24
	Reports sought for retrieval	186
	Reports not retrieved	13
Eligibility	Reports assessed in full text	173
	Excluded at full text (no parameters 55, species out of scope 3, secondary compilation 3, captivity 1)	62
Inclusion	Reports included from the search	111
	Reports of the initial database (52 before the search, minus 1 outside Brazilian waters)	51
	Total reports included (162 distinct sources, 150 primary studies, 19 apud sources)	**162**

Figure 1 presents the flow in the PRISMA 2020 format. It is generated by the script prisma_flow.py from the file prisma_flow_numbers.csv, which checks the arithmetic of each stage before drawing.



Adapted from PRISMA 2020 (Page et al. 2021, BMJ 372:n71).

Figure 1. *PRISMA 2020 flow diagram of the search, screening and inclusion of reports.*

The 52 initial-database reports correspond to studies extracted before the systematic search, from earlier compilations, and enter directly into inclusion, in the PRISMA 2020 "previous studies" box. For the included

reports, PRISMA distinguishes studies from reports: the 162 sources correspond to 150 primary studies, because theses, dissertations and derived articles describe the same data (Section 10.1).

9. Extraction, harmonization and curation

This section describes how values were transcribed from the works into the master spreadsheet and how the database was cleaned.

9.1. Recording unit and extraction rules

Each row corresponds to one estimate as published, for one sex, one estimation method and one locality. The rules below were fixed during extraction and are recorded in the change log sheet of the spreadsheet.

- The value is transcribed in the original unit and measure. Conversions are made in derived columns, by script, and not in the cell holding the raw value. The coefficient k keeps its original unit (year, day, month or bimonth) and is harmonized to $year^{-1}$ only in the derived columns.
- Relationships between total length and carapace length: length_rel_a is the intercept and length_rel_b the slope, for CL as a function of TL or the inverse. Power-form relationships receive their own unit label (TL~CL (mm; power)), and relationships with tail or abdominal length receive another (TL~TAIL (mm)); neither is comparable to the linear relationships. Weight-length relationships are included only when of power form, with the unit stated explicitly (g/mm (CL), g/mm (TL) or g/cm (TL)).
- When the work does not state whether the length is TL or CL, the measure is inferred by biological plausibility (range of observed lengths and magnitude of the parameter itself), and the inference is flagged in the row notes.
- Non-comparable measures (for example, carapace length taken from the postorbital margin) are kept, flagged and excluded from $L\infty$ comparisons.
- The value in the article prevails over any recalculation. When Z differs from F + M, when Z is smaller than M, or when F is extreme and was reported as such by the author, the published value is kept and the inconsistency is recorded in the qc_issues sheet (21 records at the date of this version).
- Evident transcription errors in the work itself (for example, a lost decimal point) are corrected only with confirmation in the text or table of the work; the correction is logged, with the old and the new value.
- When the same source appears with more than one wording of its reference, a single wording is adopted.

9.2. Procedure, independent validation and use of computational assistance

Screening and extraction were conducted independently by two researchers (M.S.S. and A.C.A.F.). Each located and read the works and transcribed the values to the spreadsheet; each then validated their own extraction with a large language model (Claude, Anthropic, on separate accounts), used for PDF reading, transcription checking, standardization of units and nomenclature, duplicate detection and flagging of inconsistencies. Finally, the two result sets were compared independently. Where the values did not match, the article was re-read to check and extract the correct information.

Agreement is reported as the number of sources with divergence. One or two of the 162 sources were divergent, and in every case the cause was the internal inconsistency of the published work itself, whose estimates do not hold across the text, the tables and the figures. These cases follow the rule of Section 9.1: the published value is kept and the inconsistency is flagged. Because the comparison covered all sources, and not a sample, 20% sampling and the calculation of Cohen's κ do not apply; κ measures classification agreement, not numerical transcription.

Eligibility decisions and the handling of doubtful cases were made by the researchers, and every relevant change is logged in the spreadsheet with the old value, the new value and the rule applied. The use of computational assistance is declared here and will be declared in the manuscript according to the journal policy.

9.3. Curation, deduplication and traceability

Deduplication of the master spreadsheet was done by row-by-row comparison within each source and between sources that reproduce the same relationship. The removed rows were not discarded: they remain in the removed_rows sheet, with the reason and the equivalent row that was kept, and the correspondence between the old and the new numbering is in the row_map sheet. All other modifications (standardization of units, vocabulary and references, transcription corrections) are in the change_log sheet, with 4,859 records, each with row, column, old value, new value and rule applied. The scripts that executed these operations are versioned and form part of the supplementary material.

Table 7. Rows removed from the master spreadsheet in the October 2026 curation, by reason.

Reason	Rows	Criterion
Exact duplicate (same source, sex, locality and values)	9	The row with the lower number is kept; the other is removed.
Partial duplicate (all content already present in a more complete row)	26	The row with more fields is kept.
Same weight-length relationship reproduced by a later source (Santos & Freitas 2005, via Santos et al. 2006)	2	The original source is kept, which gives n and period.
Captivity studies (one growth experiment and one broodstock tank experiment)	18	Ineligible for natural populations (Section 2.1).
Study outside Brazilian waters (Cienfuegos Bay, Cuba; Morenza et al. 1992)	2	Ineligible by the geographic scope (Section 2.3); overlooked during screening.
Total removed	**57**	**From 1,379 to 1,322 data rows.**

10. Reliability and non-independence among estimates

10.1. Sources of non-independence

The rows of the database are not independent observations. Six structures generate dependence and are handled in the models rather than ignored:

- **Several estimates per source.** On average, 8.2 estimates per source (1,322 rows in 162 sources), because each work reports different sexes, methods and localities.
- **The same population in more than one source.** Of the 184 populations defined, 36 were sampled by two or more sources; raw data from a dissertation, for example, reappear in articles.
- **Secondary citations.** 57 estimates (19 sources) are apud; the same value may enter through more than one secondary source.
- **Research group.** Groups use their own sampling, measurement and estimation protocols, so that group is, in part, a method effect.
- **Derived parameters.** Natural mortality estimated by empirical equations (Pauly, Taylor, Then et al., Rikhter & Efanov) and longevity 4.605/k are functions of k and L∞; they are not additional information.
- **Structural correlation between k and L∞.** The two parameters are estimated jointly from the same curve and are negatively correlated by construction.

10.2. Identifiers and control variables

The population (population_id) is defined by the combination of species, locality and period: estimates of the same species, at the same site, from sources of the same group and with more than 50% overlap in years receive the same identifier, with documented manual exceptions. Each population receives a grouping confidence level: single source (148), probable (23), confirmed (6) and inferred (6); one geographic-distribution entry, without sampling, is not applicable. The research group is assigned by a dictionary (research_groups sheet, 20 groups and two residual categories, "Independent" and "Syntheses/Government", which are not groups).

The following columns were added to the master spreadsheet on 14 October 2026; the hierarchical models cannot be closed without them. No existing value was altered, and the operation is recorded in the change_log sheet:

- estimate_id (primary key), primary_study_id (who produced the value: 150 primary studies for 162 sources), source_read_id (where the value was read) and reliability_level (scale from 1 to 5).
- publication_type, cl_definition (standard CL, postorbital CL, CL with rostrum, TL, not reported) and growth_method_class (modal progression, ELEFAN/frequency, linearization/regression, direct age, not reported/other).
- lat_dec, lon_dec and coord_precision: coordinates of the centroid of the sampled area, to model geographic variation continuously.
- repeated_parameters, analysis_use (main, sensitivity, redundant or exclude) and use_reason: they mark, at the parameter level, the values that reappear in another publication, so that the same estimate is not counted twice.

Not yet recorded: origin of the value (table, text, figure or calculated), accepted name with AphiaID, and the standard error or confidence interval of L∞, k and L50 when reported; the database has no uncertainty field, and no weighted analysis is planned (Section 11.3).

10.3. Reliability scale

Level	Description
1	Peer-reviewed primary article, with original data and method.
2	Defended thesis or dissertation, with accessible full text.
3	Institutional technical report or conference proceedings paper, with accessible text.
4	Secondary citation, with original inaccessible and value consistent among sources.
5	Secondary citation, with original inaccessible and value divergent among sources.

Pre-specified rule: the main analysis uses levels 1 to 3; levels 4 and 5 enter only in the sensitivity analysis.

10.4. Identifiability diagnostics

Regions, research groups and methods are strongly confounded in the database. Of the 20 research groups, 13 operate in a single macro-region and only seven appear in more than one; three groups (CEPENE, NEBECC-UNESP and UENF) account for 49.5% of the rows; the UENF group contributes 194 rows from only six sources. As for method, modal progression accounts for 50.8% of the 421 L∞ estimates and the ELEFAN family for 26.6%, and the choice of method varies with the group. As a consequence, a region effect estimated without control may be, to a large extent, a group or method effect.

Before the fit of each model, the following will be reported: cross-tabulations of group × region × species × method; the rank of the design matrix; the variance attributed to each random effect with its interval; and, in Bayesian models, the sensitivity to the prior distributions. Region effects will be presented as conditional associations, not as causal effects, and inferences will be restricted to the contrasts that the design allows to be estimated, chiefly those provided by the seven groups operating in more than one region and by the populations sampled by more than one source.

11. Statistical analysis plan

This plan is fixed before the models are run and is deposited in the protocol registry (OSF) as a separate document, which gives the full specification; analyses not foreseen here are identified as exploratory in the manuscript.

11.1. Questions, responses and model structure

The primary response is the growth coefficient k, standardized to year^{-1}. Because k does not depend on the length measure, it is analyzed with carapace length (CL) and total length (TL) estimates together. The growth performance index φ′ and the size at 50% maturity (L50) depend on the length measure and are analyzed separately in the CL and TL strata, without conversion between them. L∞ is described by species and length measure but is not analyzed as a response.

Table 8. Hypotheses, response variables and effects structure of each model.

Hypothesis	Response	Fixed effects	Random effects
H1. Species differ in growth; the expected direction is that they differ, largely because of the estimation method.	log k; φ′ (secondary)	Species; method class; sex; latitude	Research group, study, population
H2. Growth does not vary with latitude once research group and method are accounted for; the latitude slope is expected to be practically null (95% credible interval within ±10% change in k per 10° of latitude).	log k; φ′	Latitude (per 10°, positive southward); species; method class; sex	Research group, study, population
H3. The estimation method alters the estimated k (modal progression, ELEFAN/frequency analysis, linearization/regression, direct age); no direction is specified.	log k	Method class; species; sex; latitude	Research group, study, population
H4. Variance in growth is attributable to research group, study and population beyond the residual (variance partitioning, not a test).	log k; φ′	None (null model), then the model of H1	Research group, study, population
H5. L50 differs between sexes (females larger).	log L50 (CL; TL)	Sex; species; latitude	Research group, study, population

Two analyses are declared in advance as exploratory: (E1) the association of log k with the year of data collection, with the same fixed and random effects as H1, and (E2) the latitude slope estimated for *Xiphopenaeus kroyeri* alone. Natural mortality M and total mortality Z are not used as responses: nearly all M values in the database come from empirical equations that are functions of k, L∞, longevity or temperature (Section 11.4), and Z depends on fishing mortality.

Species enters as a fixed effect, because it has few levels (five species in the current database) to estimate a variance reliably; separate models by species are not fitted, since only *X. kroyeri* has more than 12 populations

contributing k. Research group, study and population enter as random intercepts, so that an estimate is modeled as belonging to a population, reported by a study and produced by a group. Study is nested in group and population in study, with a few exceptions, and the intercepts are therefore specified with unique identifiers at each level. All models are Bayesian multilevel models with Gaussian errors on the log scale, fitted with the same structure so that effects are comparable among hypotheses. Ordination analyses (PCA or PCoA) on population means are exploratory and will be performed only with the number of populations that have all the parameters involved, which will be reported.

11.2. Transformations and units

Positive, skewed parameters are analyzed on a logarithmic scale. The units of k are harmonized to year^{-1} (month × 12, day × 365) and lengths are analyzed in the measure in which they were reported, without conversion between TL and CL.

11.3. Inference, variance partitioning and equivalence margin

Fixed effects are reported as posterior medians with 95% highest-density credible intervals and the probability of direction. Variance partitioning (H4) uses a null model with the three random intercepts to give the proportion of variance at each level, followed by the full model, for which the Bayesian R^2 of the fixed effects and of the whole model and the change in the expected log predictive density (loo) when each fixed term is removed are reported. The contribution of the estimation method is reported as an effect size, as the variance it explains and as the change in loo, and not as a test based on an information criterion alone.

For H2, a change smaller than ±10% in k per 10° of latitude is defined in advance as biologically irrelevant (±0.095 on the natural-log scale; ±0.041 for φ′ on the log10 scale). The latitude slope is classified as practically null when its whole 95% credible interval lies inside this margin, as inconclusive when the interval crosses the margin, and as different from zero when the interval lies wholly outside it. Most estimates carry no standard error, and the master spreadsheet has no uncertainty field; all models are therefore unweighted.

11.4. Derived parameters and fixed values

M obtained by empirical equations and longevity computed as 4.605/k are not used as independent response variables alongside k and L∞. L∞ values fixed by convention from earlier work are not estimates and are excluded from the main analysis; they enter only in the sensitivity analysis, identified as such. Parameters flagged as repeated from another publication are omitted from the per-parameter analyses; when two works report different values for the same population, both are kept and the population random intercept absorbs their non-independence.

11.5. Sensitivity analyses and limitations

- Inclusion of the rows flagged as sensitivity (secondary citations of reliability levels 4 and 5 and the post-orbital length measure), and exclusion of rows with a quality-control flag.
- One estimate per population (the one with the largest n) and all estimates.
- Comparison between journal articles and grey literature.
- Exclusion, one at a time, of each research group (leave-one-group-out), and exclusion of the groups that contribute a single study.
- Alternative prior distributions for the standard deviations of the random effects.

Records of *Xiphopenaeus kroyeri* may include more than one species, because the source works do not distinguish them (taxonomic uncertainty before 2019). This heterogeneity is already present in the published estimates, cannot be resolved by the statistical analysis of the parameters, and is declared as a limitation that may inflate the residual variance of that species.

11.6. Software and reproducibility

Analyses are conducted in R, with brms (Bürkner 2017; Stan backend) for the models, the posterior and bayesplot packages for summaries and diagnostics, loo (Vehtari et al. 2017) for model comparison and emmeans for contrasts, in versioned scripts that generate all tables and figures of the manuscript. R and package versions are recorded with sessionInfo(). Conclusions are restricted to conditional observational associations and to variance partitioning, and the heterogeneity expected among studies is presented as a result, not as noise.

12. Current state of the extraction database

Section 8 reports the screening of the search; this section describes the master spreadsheet of extraction on 14 October 2026, after the curation of Section 9.3. It contains 1,322 estimates, extracted from 162 sources, referring

to 184 populations and 22 research-group categories (20 groups plus the two residual categories "Independent" and "Syntheses/Government"), with start years of data collection between 1958 and 2025.

Table 9. Estimates, sources and populations by species. Sources and populations may cover more than one species.

Species	Estimates	Sources	Populations
Penaeus (Xiphopenaeus) kroyeri	634	93	92
Penaeus (Litopenaeus) schmitti	263	44	37
Penaeus (Farfantepenaeus) subtilis	239	36	27
Penaeus (Farfantepenaeus) paulensis	107	20	16
Penaeus (Farfantepenaeus) brasiliensis	78	16	15
Farfantepenaeus spp. (not identified to species level)	1	1	1
Total (distinct sources and populations)	**1,322**	**162**	**184**

Table 10. Number of estimates (rows) with at least one reported value, by parameter.

Parameter	Rows	Parameter	Rows
L∞ (TL or CL)	421	TL~CL relationship (length_rel_a, length_rel_b)	97
Growth coefficient k	365	Weight–length relationship (wl_a, wl_b)	200
Parameter t_0	189	Asymptotic weight W∞	31
Longevity (tmax)	264	L50 (CL or TL)	144
Natural mortality M	211	SL50 (CL or TL)	52
Total mortality Z	207	Recruitment length (Lr)	54
Fishing mortality F	173	Reproductive period	163
Recruitment period	196		

Table 11. Estimates by macro-region and number of research-group categories with data in each.

Macro-region	Estimates	Group categories
Southeast	541	10
Northeast	491	11
North	162	4
South	87	6
Southeast–South (continuous area)	24	2
No region assigned	17	1
Total	**1,322**	**—**

Table 12. Estimates, sources and macro-regions of activity by research group.

Group	Estimates	Sources	Macro-regions
CEPENE-NE	252	32	NE, SE
NEBECC-UNESP	209	23	S, SE
UENF-RJ	194	6	SE
UFPA-N	88	7	N, NE
UFS-Hirose	81	4	NE
UFRPE-PE	77	10	NE, SE
Independent	73	9	NE, SE, SE–S, no region
UFRA-PA	69	7	N
IP-SP	53	13	SE
UNIVALI-SC	37	10	S
UFS-Freire	32	6	NE, SE
UFRRJ-RJ	27	2	SE
FURG-RS	27	6	S
Syntheses/Government	26	8	N, S, SE–S, no region
UFPB-PB	15	2	NE
UFES-ES	15	4	NE, SE
UFPR-PR	9	3	S
UFC-CE	8	2	N, NE

Group	Estimates	Sources	Macro-regions
UNESP-RC	8	2	SE
UFSC-SC	8	2	S
UFBA-BA	8	2	NE
UFMA-MA	6	2	NE

The coverage is very uneven: *X. kroyeri* accounts for 48.0% of the estimates and *F. brasiliensis* for 5.9%; the Southeast and Northeast together account for 78.1% of the rows. As to quality, 21 records are flagged in the qc_issues sheet (values that are strange but reported by the authors, relationships of a different form, a non-comparable measure, and internal inconsistencies of the article itself), and none was corrected without confirmation in the original work.

13. Limitations

The limitations are declared in advance and delimited; each is accompanied by the measure that resolves it or circumscribes its effect.

- **Sources not searched.** Zoological Record, BIOSIS Previews, LILACS and Google Scholar were not searched; SciELO Brasil and BDTD/IBICT were covered only indirectly through Crossref, DOAJ and institutional repositories (Section 3.2). Coverage of old grey literature depends on citation searching and institutional repositories.
- **Counts of the complementary search.** The interfaces used in the assisted retrieval do not provide stable exportable counts per query; the complementary-search count corresponds to the unique records of the consolidated list plus records excluded before consolidation and duplicates, less the database, citation and grey-literature counts (Section 4.3). Counts of the screening and inclusion stages are exact.
- **Sources with blocked automated access.** SciELO Search, BDTD/IBICT, CAPES Catalog of Theses, AGRIS/FAO, Google Scholar, oasisbr, Revista CEPSUL, Boletim Técnico-Científico do CEPNOR and Acta of Fisheries and Aquatic Resources blocked programmatic queries; the repositories of UFRA, UFRN and UFRJ had technical failures (Appendix A). Mitigation: direct consultation of the interfaces by the researchers (database and citation searches, Section 3.1).
- **CEPENE bulletin volumes without a text layer.** Volumes 1–11, 13(1) and 15(1) of the Boletim Técnico-Científico do CEPENE are scanned without optical character recognition and cannot be reached by text search; since this series concentrates the Northeast production on the four species, the gap is systematic and directional, not random; 13 reports remained not retrieved in the final count. Mitigation: page-by-page scanning of the issue tables of contents.
- **Manual and non-exhaustive citation searching.** Thirty records were identified by direct and reverse citation searching. Mitigation: complement with the "cited by" functions of Web of Science and Scopus in a search update.
- **Agreement between researchers without κ.** Screening and extraction were carried out by two researchers, and agreement is reported as the number of divergent sources, not as Cohen's κ (Sections 5.3 and 9.2).
- **Taxonomic uncertainty before 2019.** *Xiphopenaeus dincao* was described in 2019; earlier records of "*X. kroyeri*" from the Southeast and South of Brazil may correspond to that species or to a mixture of the two. Mitigation: the name as published, the accepted name and the AphiaID are recorded in separate fields, and a sensitivity analysis by latitude is conducted. The same applies to the possible cryptic species in the *F. subtilis* complex recorded in the literature.
- **Extraction with language-model assistance.** Mitigation: extraction by two researchers with independent comparison of all sources, reporting of the number of divergences and declaration of the use of computational assistance (Section 9.2).
- **Confounding among research group, region and estimation method.** Mitigation: crossed random effects, identifiability diagnostics and interpretation restricted to estimable contrasts (Section 10.4).
- **Heterogeneity in length definitions** (TL, standard CL, post-orbital CL, CL with rostrum) and inference of the measure when the work does not state it. Mitigation: a measure-definition column and stratified analysis (Section 9.1).
- **Derived estimates and secondary citations can inflate apparent replication.** Mitigation: primary-study identifiers, a reliability scale and sensitivity analyses (Sections 10 and 11.5).

- **No standard error for most estimates,** which prevents weighting. Mitigation: all models are unweighted, and the wide intervals expected from the small number of populations and groups are reported as such (Sections 10.4 and 11.3).
- **No rows attributed to *X. dincao*** in the current database. Mitigation: targeted search for post-2019 studies of the genus and registration with the accepted name (Section 2.1).

14. Reproducibility and availability

The following are made available as supplementary material with a persistent identifier: (i) the registered protocol, with the true date of registration; (ii) the complete list of the 188 records screened, with priority class, persistent identifier, species, expected parameters and status; (iii) the search log of Appendix A, with source, access mode and outcome; (iv) the named list of records excluded at full text, with categorized reason; (v) the data dictionary of the extraction spreadsheet, with field name, type, unit and controlled vocabulary; (vi) the file of extraction decisions, versioned and dated; (vii) the PRISMA 2020, PRISMA-EcoEvo and PRISMA-S checklists, indicating the section in which each item is met; (viii) the code for cleaning, conversion and analysis, including the scripts that organized and deduplicated the master spreadsheet; and (ix) the statistical analysis plan, deposited before the models are run (Section 11).

The database will be deposited with a DOI and a CC-BY license, and the PRISMA flow diagram is generated by script from the bookkeeping spreadsheets, not drawn by hand, so the numbers in the diagram and those in the tables are necessarily identical.

Methodological references

Bürkner, P.-C. 2017. brms: an R package for Bayesian multilevel models using Stan. Journal of Statistical Software 80(1): 1-28.

CEE — Collaboration for Environmental Evidence. 2022. Guidelines and Standards for Evidence Synthesis in Environmental Management, version 5.0. Pullin, A.S.; Frampton, G.K.; Livoreil, B.; Petrokofsky, G. (eds.).

Gusenbauer, M.; Haddaway, N.R. 2020. Which academic search systems are suitable for systematic reviews or meta-analyses? Research Synthesis Methods 11: 181-217.

Haddaway, N.R.; Macura, B.; Whaley, P.; Pullin, A.S. 2018. ROSES Reporting standards for Systematic Evidence Syntheses. Environmental Evidence 7: 7.

James, K.L.; Randall, N.P.; Haddaway, N.R. 2016. A methodology for systematic mapping in environmental sciences. Environmental Evidence 5: 7.

Kruschke, J.K. 2018. Rejecting or accepting parameter values in Bayesian estimation. Advances in Methods and Practices in Psychological Science 1(2): 270-280.

Livoreil, B. et al. 2017. Systematic searching for environmental evidence using multiple tools and sources. Environmental Evidence 6: 23.

Nakagawa, S.; Schielzeth, H. 2013. A general and simple method for obtaining R^2 from generalized linear mixed-effects models. Methods in Ecology and Evolution 4(2): 133-142.

O'Dea, R.E. et al. 2021. Preferred reporting items for systematic reviews and meta-analyses in ecology and evolutionary biology: a PRISMA extension. Biological Reviews 96: 1695-1722.

Page, M.J. et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372: n71.

Pauly, D.; Munro, J.L. 1984. Once more on the comparison of growth in fish and invertebrates. Fishbyte 2(1): 1-21.

Rethlefsen, M.L. et al. 2021. PRISMA-S: an extension to the PRISMA statement for reporting literature searches in systematic reviews. Systematic Reviews 10: 39.

Schneider, F.D. et al. 2019. Towards an ecological trait-data standard. Methods in Ecology and Evolution 10: 2006-2019.

Thorson, J.T. et al. 2017. Predicting life history parameters for all fishes worldwide. Ecological Applications 27: 2262-2276.

Vehtari, A.; Gelman, A.; Gabry, J. 2017. Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. Statistics and Computing 27: 1413-1432.

WoRMS Editorial Board. 2026. World Register of Marine Species. Accessed September 2026.

Appendix A — Search log

Sources consulted on 7 and 8 September 2026 by AI-assisted programmatic retrieval and direct navigation (updated 12–13 October 2026), with access mode and outcome. The record of inaccessible sources is an integral part of the log.

Source consulted	Access mode	Outcome	Note
Crossref REST API	endpoint /works, query.bibliographic, 40 records per query	Worked, with rate limiting after successive queries	Main structured index. Covers SciELO Brasil, Bol. Inst. Pesca, Nauplius, Arq. Ciênc. Mar, Atlântica, Rev. Bras. Zool. and USP theses
DOAJ API	endpoint /api/search/articles	Worked	High yield for Brazilian open-access journals
OpenAlex API	filter=title_and_abstract.search	Failed — HTTP 429 on all attempts	Rate limit of the network address used, not of the search scope
OJS portal of Bol. Inst. Pesca	navigation and search of the index	Worked — second-largest source of yield	The system assigns the year 2018 to almost all retro-digitized articles; the year is read from the body of the record
CEPENE/ICMBio portal	issue and article PDFs	Partly worked	Vols. 12, 13(2), 14 (partial), 15(2), 16, 17 and 18 with extractable text; vols. 1-11, 13(1) and 15(1) without OCR or incomplete
Institutional repositories (USP, UNESP, UFPA, UFPB, UFAL, UFRRJ, UFS, UFPR, FURG, UENF)	web search restricted by domain and direct access to the bitstream	Partly worked	The DSpace search endpoints block automated access; individual files open normally
General web search	search engine	Worked	Workaround route for grey literature and for sources with blocked programmatic access
SciELO Search	—	Blocked by robots.txt	Bypassed through Crossref and DOAJ, which index SciELO Brasil. Not accessible to the assisted retrieval
BDTD/IBICT	—	Blocked by robots.txt	Bypassed through search of the institutional repositories. One UFES record visible only through BDTD remains not retrieved
Catálogo de Teses e Dissertações da CAPES	—	Blocked — JavaScript-only interface	Not accessible to the assisted retrieval
AGRIS/FAO	—	Blocked by robots.txt	Not accessible to the assisted retrieval
Google Scholar	—	Blocked by robots.txt	Not accessible to the assisted retrieval; stopping rule declared
oasisbr / IBICT	—	Blocked by robots.txt	Not accessible to the assisted retrieval
Revista CEPSUL — Biodiversidade e Conservação Marinha	—	Blocked by robots.txt	Records identified by indirect web search
Bol. Téc.-Cient. CEPNOR	—	Blocked by robots.txt	Records identified by web search and by DOI
Acta of Fisheries and Aquatic Resources	—	Blocked by robots.txt	Not accessible to the assisted retrieval
REVIZEE reports	—	Blocked by robots.txt	Not accessible to the assisted retrieval
UFRA repository	—	Failed — invalid digital certificate	Two records on the closed season (defeso) of *F. subtilis* on the north coast remain unverified
UFRN repository	—	Failed — HTTP 403	Not accessible to the assisted retrieval
Pantheon/UFRJ	—	Failed — connection error	One record of the Museu Nacional remains unverified

Appendix B — Records not retrieved

Twenty-four records had their existence verified but their full text was not located online at the September 2026 consolidation. They are not exclusions: they were to be resolved by physical collection consultation, by interlibrary loan request or by contact with the authors. Their predominant origin is the CEPENE Boletim Técnico-Científico series, in the volumes scanned without optical character recognition, a systematic and geographically directed gap concentrated in Northeast production.

This appendix reproduces the list of pending records of the September consolidation; the current status of each record is in the records sheet of the search workbook; in the final count (Table 6) 13 reports remained not retrieved.

Three of these records have a methodological importance that exceeds their data content, as they are the original sources of criteria applied recurrently by the Brazilian literature and reproduced, by secondary route, in dozens of rows of the database: D'Incao & Fonseca (1998/1999), the origin of the longevity criterion equal to 4.605/k; Zenker

& Agnes (1977), the origin of L∞ values adopted as fixed by later works; and Mello (1973), needed to settle a divergence in sex assignment between two included works.

Table B1. Records not retrieved, with outlet and reason not located.

Authors and year	Title	Outlet	Reason not located
ZENKER, H.H.; AGNES, J.L. 1977	Distribuição do camarão-rosa (Penaeus brasiliensis e Penaeus paulensis) ao longo da costa sudeste e sul do Brasil	Série Documentos Técnicos, SUDEPE-PDP, 21: 1-105	Source of the 'fixed' CL∞ of Salvati et al. (2021) and Peixoto et al. (2001); reports a latitudinal recruitment gradient; 105 pages; probably held only in physical collections (libraries of IBAMA/CEPSUL, USP and FURG).
MELLO, J.T.C. 1973	Estudo populacional do camarão-rosa Penaeus brasiliensis e Penaeus paulensis	Bol. Inst. Pesca 2: 19-65	Resolves the sex-assignment conflict between Peixoto et al. (2001) and Perroca et al. (2022). The OJS of Bol. Inst. Pesca covers only retro-digitized issues from v.15 onward.
D'INCAO, F. 1984	Estudos sobre o crescimento de Penaeus paulensis na Lagoa dos Patos, Rio Grande do Sul	Atlântica 7: 73-84	Cited by Peixoto et al. (2001), Rezende et al. (2015) and Salvati et al. (2021). The Atlântica (FURG) portal does not expose the old volumes.
D'INCAO, F.; FONSECA, D.B. 1998 or 1999	Performance of the von Bertalanffy growth curve in penaeid shrimps: a critical approach	Proceedings of the Fourth International Crustacean Congress, p. 733-737, Amsterdam	Source of the 4.605/k criterion used in dozens of rows of the database and not consulted in the original. Peixoto et al. (2001) list it as 1998; Miazaki et al. (2021) and Perroca et al. (2022) cite 1999. The year remains unresolved pending retrieval.
SANTOS, M.C.F.; IVO, C.T.C. 2000	Pesca, biologia e dinâmica populacional do camarão sete-barbas capturado em frente ao município de Caravelas (Bahia)	Bol. Téc.-Cient. CEPENE 8(1): 131-164	The PDF btc-vol08.pdf is online but is a scan without OCR. Longevity 2.30/2.70 years by an undeclared criterion.
SANTOS, M.C.F. 2000	Biologia e pesca de camarões marinhos ao largo de Maragogi (Alagoas)	Bol. Téc.-Cient. CEPENE 8(1): 99-129	Same volume, scanned without OCR.
SANTOS, M.C.F.; FREITAS, A.E.T.S. 2000	Pesca e biologia dos peneídeos capturados no município de Barra de Santo Antônio (Alagoas)	Bol. Téc.-Cient. CEPENE 8(1): 73-98	Same volume, scanned without OCR.
SANTOS, M.C.F.; FREITAS, A.E.T.S. 2002	Camarões marinhos capturados com arrastão de praia e arrasto motorizado ao largo de Pitimbu (Paraíba)	Bol. Téc.-Cient. CEPENE 10(1): 145-170	The PDF btc-vol10.pdf is online, scanned without OCR.
SANTOS, M.C.F.; FREITAS, A.E.T.S.; MAGALHÃES, J.A.D. 2003	Aspectos biológicos do camarão sete-barbas capturado ao largo do município de Ilhéus (Bahia)	Bol. Téc.-Cient. CEPENE 11(1): 175-187	The PDF of vol. 11 on the ICMBio portal contains only one article, not the complete issue.
SANTOS, M.C.F.; FREITAS, A.E.T.S. 2006	Caracterização biológica e pesqueira do camarão sete-barbas no pesqueiro Laminha, APA de Piaçabuçu (Alagoas)	Bol. Téc.-Cient. CEPENE 14(1): 71-92	The portal exposes only article 09 of vol. 14(1).
SANTOS, M.C.F.; FREITAS, A.E.T.S. 2006	Determinação biológico-pesqueira do camarão sete-barbas em Jequiá da Praia (Alagoas)	Bol. Téc.-Cient. CEPENE 14(2): 61-71	The portal exposes only article 03 of vol. 14(2).
SILVA, U.M.; SANTOS, M.C.F. 2006	Estrutura populacional do camarão sete-barbas no município de Passo de Camaragibe (Alagoas)	Bol. Téc.-Cient. CEPENE 14(2): 51-60	Same as above.
SANTOS, M.C.F.; FREITAS, A.E.T.S. 2007	Avaliação biológica dos camarões peneídeos capturados no município de São José da Coroa Grande (Pernambuco)	Bol. Téc.-Cient. CEPENE 15(1): 63-79	The complete volume 15(1) is online at https://www.icmbio.gov.br/cepene/images/stories/publicacoes/btc/vol15/btc-vol15.pdf.
NATIVIDADE, C.D. 2006	Estrutura populacional e distribuição do camarão sete-barbas no litoral do Paraná	Dissertação de mestrado, UFPR	Handle located: https://acervodigital.ufpr.br/handle/1884/5203. The portal blocks automated access but opens in a browser.
SANTOS, M.C.F. 1997	O camarão sete-barbas, Xiphopenaeus kroyeri, no Nordeste do Brasil	Dissertação de mestrado, UFPE, 232 p.	Source of several series of percentages of immature females used as secondary citations (apud) in the database.
CORRÊA, J.; MARTINELLI, J.M. 2009	(penaeid shrimps)	Rev. Cient. UFPA 7(1): 1-18	Reference confirmed; no online version located.

Authors and year	Title	Outlet	Reason not located
ISAAC, V.J.; DIAS-NETO, J.; DAMASCENO, F.G. 1992	Camarão-rosa da costa norte: biologia, dinâmica e administração pesqueira	IBAMA, Série Estudos de Pesca 1	Classic technical document of the north coast.
D'INCAO, F.; CALAZANS, D.K. 1978	(penaeid shrimps)	(to be confirmed)	Outlet not confirmed.
CINTRA, I.H.A. et al. 2004	(north-coast pink shrimp)	Bol. Téc. Cient. CEPNOR 4(1): 21-29	The CEPNOR portal blocks automated access.
NASCIMENTO, P.A.M.; POLI, C.R. 1985	(seabob shrimp)	(to be confirmed)	Outlet not confirmed.
SANTOS, M.C.F. et al. 2006	(penaeid shrimps)	Bol. Téc.-Cient. CEPENE 14 — pagination conflicting between sources: 14(1): 59-70 or 14: 33-58	Pagination conflict unresolved.
(to be identified) [n.d.]	Análise bio-econômica dos defesos do camarão-rosa (Penaeus subtilis) na costa norte do Brasil	Repositório UFRA, handle 123456789/2294	Probably yield per recruit and exploitation rates. The UFRA repository has an invalid SSL certificate and does not open by automated access.
(to be identified) [n.d.]	Biologia reprodutiva do camarão sete-barbas capturado no litoral sul do Espírito Santo	Dissertação, UFES (record seen in BDTD)	BDTD record: UFES_a245e2bf0bd212a7148e196068fb7ccf. Handle in the UFES repository not located.
(to be identified) [n.d.]	Aspectos bioecológicos e pesqueiros de três espécies de camarões do gênero Penaeus nas costas do estado do Rio de Janeiro e experimentos de cultivo	UFRJ, Museu Nacional, handle 11422/2677	The Pantheon/UFRJ page did not open; metadata not retrieved.

Appendix C — List of sources and outcomes

The complete list of the 188 records of the assisted-retrieval list, with authors, title, outlet, persistent identifier, species, expected parameters, priority class and status, is provided in an accompanying spreadsheet (file busca_2026_lista_referencias.xlsx), organized in three sheets: records, search_log and summary (plus web_search_candidates). The spreadsheet is the supplementary material of reference for this document and is deposited with the database.

Appendix D — Search strategy per database and execution record

This appendix meets PRISMA-S: for each database, the exact string, the execution date, the coverage and the number of records returned. The same search string was used in all databases (structure of Section 4), adapted only to the syntax of each interface.

Table D1. Adaptation of the strategy to each database.

Database	Fields searched	Syntax adopted
Web of Science Core Collection	TS= (title, abstract, author keywords and Keywords Plus)	TS=((kroyeri OR dincao OR subtilis OR brasiliensis OR paulensis OR schmitti OR Xiphopenaeus OR Farfantepenaeus OR Litopenaeus OR Penaeidae OR penaeid* OR shrimp* OR prawn* OR seabob OR "sea-bob" OR "sete-barbas" OR camaron* OR "camarão" OR "camarões") AND ("von Bertalanffy" OR "growth parameter*" OR "asymptotic length" OR "L-infinity" OR ELEFAN OR FiSAT OR TropFishR OR "length-frequency" OR "natural mortality" OR "total mortality" OR "fishing mortality" OR "exploitation rate" OR "population dynamic*" OR "length at maturity" OR "size at maturity" OR "length-weight" OR recruitment OR longevity OR fecundity OR mortalidade OR crescimento OR maturidade OR recrutamento)) NOT TS=("Bacillus subtilis")
Scopus	TITLE-ABS-KEY	Same structure A AND B, with the operator TITLE-ABS-KEY( ), and the exclusion line AND NOT TITLE-ABS-KEY("Bacillus subtilis") to eliminate the noise from the epithet subtilis.
ASFA (ProQuest)	NOFT( ) or title, abstract and descriptor fields	Same structure A AND B. The interface and syntax actually used are recorded, as they vary between providers.
CAPES Periodicals Portal	Advanced search (all fields)	Same structure A AND B. The collection or provider actually consulted is recorded.
SciELO Brasil and LILACS (not searched)	All indexes	Not searched. Interfaces with limited operators; the planned approach was pairs (one term of Block A with one of Block B), in Portuguese, English and Spanish, with each pair recorded.
Google Scholar (not searched)	Full text	Not searched. The planned approach was term pairs with a stopping rule: the first 200 references by relevance, screened by title. It is a complementary source, not a principal one (Gusenbauer & Haddaway 2020).

Table D2. Execution record of the database searches.

Database	Date	Version / years covered	Records returned	Relevant (title and abstract)
Web of Science	Jul–Oct 2026	Core Collection; inception–Oct 2026	24	20
Scopus	Jul–Oct 2026	inception–Oct 2026	36	30
ASFA	Jul–Oct 2026	inception–Oct 2026	6	5
CAPES Periodicals Portal	Jul–Oct 2026	inception–Oct 2026	6	5
Total, databases			**72**	**60**

The totals of Table D2 feed the identification box of the PRISMA flow diagram (Figure 1), generated by the script prisma_flow.py from the file prisma_flow_numbers.csv, which checks the arithmetic before drawing.